

Krishnendu Sinha

Associate Professor of Zoology

Department of Zoology, Siksha Bhavana, Visva-Bharati, Santiniketan-731235, India

Email dr.krishnendusinha@gmail.com
ORCID [0000-0001-9590-3875](https://orcid.org/0000-0001-9590-3875)
Google Scholar scholar.google.com/citations?user=ZEI-IzMAAAAJ
Scopus Author ID: 55612440000
Lab website [Sinha Molecular Evolution Group](#)
Last updated 23 June 2026

Professional Profile

Molecular evolution and computational bioinformatics researcher working at the interface of sequence analysis, adaptive molecular evolution, phylogenetics, protein evolution, and reproducible computational workflows. Current research emphasizes learned scoring for multiple sequence alignment, codon-aware analysis, episodic selection, robustness-aware evolutionary inference, and interpretable computational biology.

Academic Appointments

2026–present	Associate Professor of Zoology , Department of Zoology, Siksha Bhavana, Visva-Bharati, Santiniketan.
2015–2026	West Bengal Education Service , Jhargram Raj College, Jhargram, West Bengal.

Education

2011–2017	Ph.D. in Molecular Medicine , Bose Institute, Kolkata.
2009–2011	M.Sc. , University of Calcutta, Kolkata.
2006–2009	B.Sc. , Presidency College, Kolkata.

Research Interests

- Molecular evolution, adaptive selection, and branch-site evolutionary inference.
- Multiple sequence alignment, codon-aware alignment, and sequence-analysis methods.
- Protein evolution, structural interpretation of selection signals, and molecular function.
- Reproducible bioinformatics workflows, robustness summaries, and sensitivity-aware interpretation.
- Computational toxicology, gut microbiome signals, and neurodegeneration-associated molecular data analysis.

Software and Computational Resources

- **BABAPPAlign**: progressive multiple sequence alignment engine using learned residue-level scoring over fixed protein-language-model embeddings, with exact affine-gap dynamic programming and codon-aware alignment mode. Published in *Bioinformatics*.

- **BABAPPASnake**: workflow for orthogroup-centered episodic selection analysis with multi-engine alignment pathways, phylogenetic inference, branch-site follow-up testing, and robustness-aware summaries.

Publications and Scholarly Outputs

- [1] Lineage-specific selection signals in the Growth arrest-specific protein 8 (GAS8) domain protein of *Trypanosoma melophagium*. *Scientific Reports*. 2026. doi:10.1038/s41598-026-56233-x.
- [2] Curli Carrier Burden: a quantitative trait-level microbiome index for amyloidogenic bacterial signals in Parkinson's disease gut metagenomes. *Scholarly output*. 2026. doi:10.64898/2026.05.25.727557.
- [3] NeuroFate: endpoint-locked transcriptomic axis scoring for neurodegeneration risk research. *Scholarly output*. 2026. doi:10.64898/2026.05.23.727380.
- [4] BABAPPAlign: A Multiple Sequence Alignment Engine with a Learned Residue-Level Scoring Function. *Bioinformatics*. 2026. doi:10.1093/bioinformatics/btag189.
- [5] Episodic positive selection structurally stabilizes the Arabidopsis CONSTANS-like gene COL5 indicating adaptive evolution. *Scientific Reports*. 2026. doi:10.1038/s41598-025-34129-6.
- [6] BABAPPASnake: a workflow for episodic selection analysis with robustness-aware summaries. *bioRxiv*. 2025/2026 versioned preprint. doi:10.1101/2025.04.27.650835.
- [7] Curli protein: A potential contributor to alpha-synucleopathy in Parkinson's disease. *Translational Medicine of Aging*. 2025. doi:10.1016/j.tma.2025.02.002.
- [8] Selection on synonymous codon usage in soybean (*Glycine max*) WRKY genes. *Scientific Reports*. 2024. doi:10.1038/s41598-024-77156-5.
- [9] Pesticides and the Gut Microbiota: Implications for Parkinson's Disease. *Chemical Research in Toxicology*. 2024. doi:10.1021/acs.chemrestox.4c00057.
- [10] A Review on the Recent Applications of Deep Learning in Predictive Drug Toxicological Studies. *Chemical Research in Toxicology*. 2023. doi:10.1021/acs.chemrestox.2c00375.
- [11] Machine Learning in Drug Metabolism Study. *Current Drug Metabolism*. 2022. doi:10.2174/1389200224666221227094144.
- [12] Lupeol alters viability of SK-RC-45 (Renal cell carcinoma cell line) by modulating its mitochondrial dynamics. *Heliyon*. 2019. doi:10.1016/j.heliyon.2019.e02107.
- [13] Sulphur dioxide ameliorates colitis related pathophysiology and inflammation. *Toxicology*. 2019. doi:10.1016/j.tox.2018.11.010.
- [14] Phytochemicals and human health. *Functional Food and Human Health*. 2018. doi:10.1007/978-981-13-1123-9_10.
- [15] Unfolding the mechanism of cisplatin induced pathophysiology in spleen and its amelioration by carnosine. *Chemico-Biological Interactions*. 2018. doi:10.1016/j.cbi.2017.11.019.
- [16] Beneficial role of antioxidant molecules with therapeutic potential in cardiac disease. *Cardioprotective Natural Products: Promises and Hopes*. 2017. doi:10.1142/9789813231160_0007.
- [17] Formalin in fish trading: An inefficient practice for sustaining fish quality. *Archives of Polish Fisheries*. 2017. doi:10.1515/aopf-2017-0005.
- [18] Unfolding the mechanism of proteases in pathophysiology of gastrointestinal diseases. *Pathophysiological Aspects of Proteases*. 2017. doi:10.1007/978-981-10-6141-7_24.

- [19] Morin and its role in chronic diseases. *Advances in Experimental Medicine and Biology*. 2016. doi:10.1007/978-3-319-41334-1_19.
- [20] Taurine protects cisplatin induced cardiotoxicity by modulating inflammatory and endoplasmic reticulum stress responses. *BioFactors*. 2016. doi:10.1002/biof.1301.
- [21] Morin protects gastric mucosa from nonsteroidal anti-inflammatory drug, indomethacin induced inflammatory damage and apoptosis by modulating NF-kB pathway. *Biochimica et Biophysica Acta - General Subjects*. 2015. doi:10.1016/j.bbagen.2015.01.008.
- [22] Cadmium (Cd^{2+}) exposure differentially elicits both cell proliferation and cell death related responses in SK-RC-45. *Toxicology in Vitro*. 2014. doi:10.1016/j.tiv.2013.11.011.
- [23] Cytochrome P450s: Mechanisms and biological implications in drug metabolism and its interaction with oxidative stress. *Current Drug Metabolism*. 2014. doi:10.2174/1389200215666141125121659.
- [24] An update on oxidative stress-mediated organ pathophysiology. *Food and Chemical Toxicology*. 2013. doi:10.1016/j.fct.2013.09.026.
- [25] Oxidative stress: The mitochondria-dependent and mitochondria-independent pathways of apoptosis. *Archives of Toxicology*. 2013. doi:10.1007/s00204-013-1034-4.

Research Identity and Profiles

- ORCID: 0000-0001-9590-3875
- Google Scholar: ZEI-IzMAAAAJ
- Scopus Author ID: 55612440000
- Lab website: Sinha Molecular Evolution Group